A Comparative Analysis of Two Capacitor-Less LDO Topologies in 130nm CMOS Technology

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AVLSI 2026

Outline

1. Introduction
2. Background
3. 5T OTA-Based LDO
4. Telescopic OTA-Based LDO
5. Comparison
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Introduction

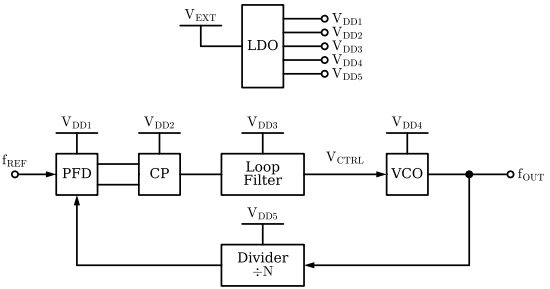


Figure: Phase Locked Loop with LDO

Parameter	Target Value
Power Supply Voltage (V_{EXT})	3 V
Bias Current (I_{bias})	5 µA
Load Current (I_L)	500 µA
Load Capacitance (C_L)	50 pF
Reference Voltage (V_{REF})	1.2 V ± 1 %
Output Voltages (V_{OUT})	2.3 V, 2.1 V, 1.6 V
Compensation Capacitor (C_c)	$<$ 7 pF
Phase Margin @ No-Load	$>$ 60°
Phase Margin @ Full-Load	$>$ 70°
Loop Gain @ 1 Hz	$>$ 50 dB
PSRR @ 1 Hz	$>$ 65 dB
PSRR @ 1 MHz	$>$ 2 dB
PSRR @ 10 MHz	$>$ 2 dB
Output-Referred Noise @ 10 Hz	< -115 dBV/$\sqrt{\text{Hz}}$
Output-Referred Noise @ 1 kHz	< -125 dBV/$\sqrt{\text{Hz}}$
Output-Referred Noise @ 10 kHz	< -134 dBV/$\sqrt{\text{Hz}}$
Output-Referred Noise @ 100 kHz	< -143 dBV/$\sqrt{\text{Hz}}$
Output-Referred Noise @ 1 MHz	< -155 dBV/$\sqrt{\text{Hz}}$

Table: Technical LDO Target Specifications

PMOS LDO Schematic

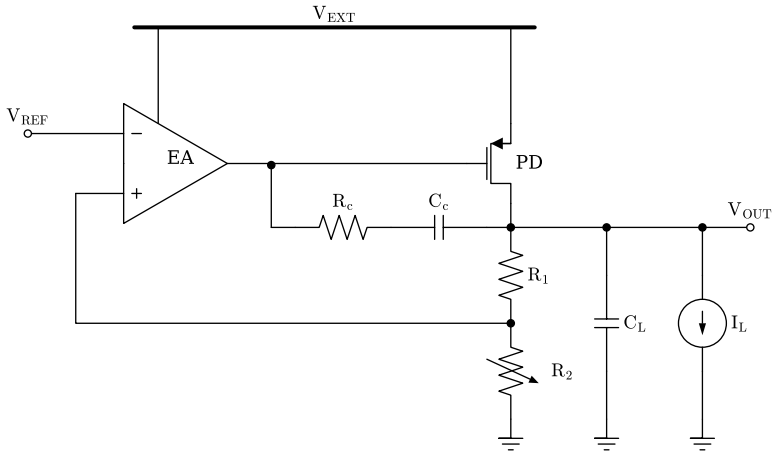


Figure: General LDO schematic

LDO's Small-Signal Model

Output voltage

$$\frac{V_{OUT}}{V_{REF}} \approx \frac{1}{\beta}, \qquad \frac{R_1}{R_2} = \frac{V_{REF}}{V_{OUT}} - 1$$

Stability

$$\omega_{p1} = \frac{1}{g_{m,PD} R_1 R_2 C_c}, \quad \omega_{p2} = \frac{g_{m,PD}}{C_1 + C_2}, \quad \omega_z = \frac{1}{C_c(1/g_{m,PD} - R_c)}$$

PSRR

$$|PSRR_{DC}| \cong \frac{1}{\beta A_{EA,0} g_{m,PD} r_{ds,p}}$$

Output-referred noise

$$S_{n,o}(f) \cong (S_{n,ref}(f) + S_{n,EA}(f))(1 + \frac{R_1}{R_2})^2 + S_{n,R2}(f)(\frac{R_1}{R_2})^2 + S_{n,R_1}(f)$$

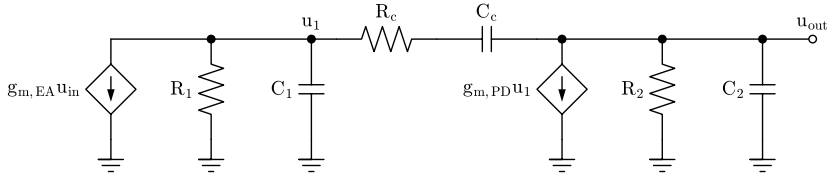


Figure: Small-signal model of the LDO

5T OTA (Simple Differential Pair)

5T OTA is a simple differential pair with a gain of:

$$A_{5T} = \frac{g_{m,12}}{g_{5T,out}}$$

whereas the output impedance:

$$g_{5T,out} = g_{ds,2} + g_{ds,4}$$

as for the output swing, in order to keep all the transistors in saturation, we must have:

$$V_{REF} - V_T \leq V_{OUT} \leq V_{EXT} - V_{DSAT}$$

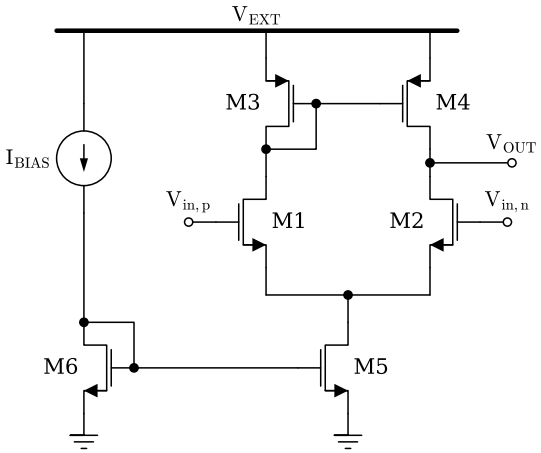


Figure: 5T OTA

5T OTA (Sizing)

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	42.0/5.0	µm
M_3, M_4 (Active Load)	3.8/5.0	µm
M_5 (Tail Current)	12.0/5.0	µm
M_6 (Input Current)	5.0/5.0	µm
M_{PD} (Pass Device)	20.0/0.14	µm
R_1 (Divider)	15.00	kΩ
R_{21} (Divider)	16.36	kΩ
R_{22} (Divider)	20.00	kΩ
R_{23} (Divider)	45.00	kΩ
R_c (Compensation)	15.00	kΩ
C_c (Compensation)	7.00	pF

Table: Device Sizing for the 5T OTA LDO

- **Input Pair:** Large W and L for reducing the Flicker noise and increasing the loop gain.
- **Active Load:** Large L for increasing the loop gain, small W for improving PSRR.
- **Tail Current:** Large L for better current mirroring, large W for increasing the bias current and reducing the Thermal noise at higher frequencies.
- **Pass Device:** Large W to improve transconductance and loop gain.
- **Divider:** Small resistors to minimize noise and decrease PD's gate voltage @ No-Load.
- **Compensation:** Nulling resistor and compensation capacitor for improving UGBW and PSRR at higher frequencies.

5T OTA (Simulation Results)

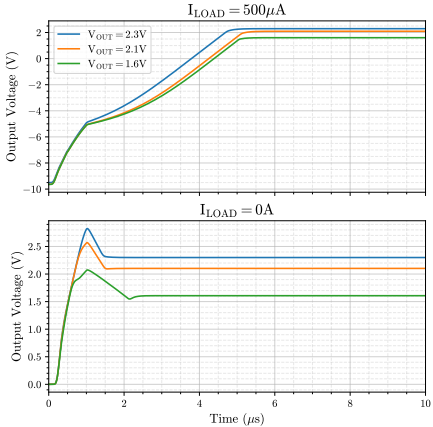


Figure: Transient Response

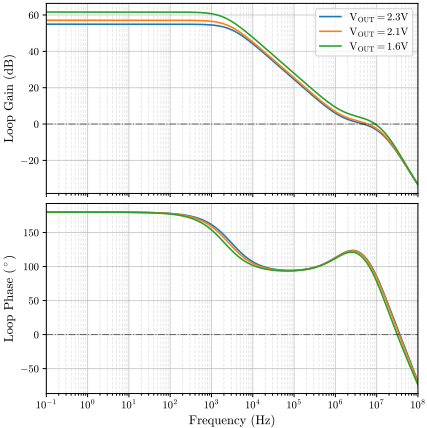


Figure: Bode Plot

5T OTA (Simulation Results)

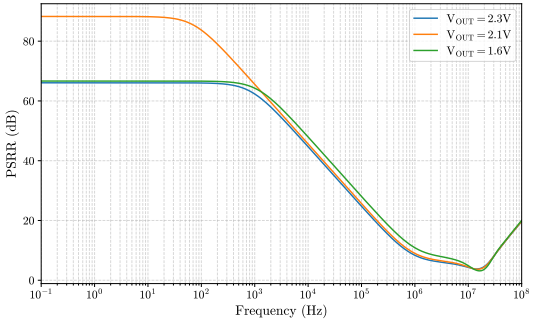


Figure: PSRR

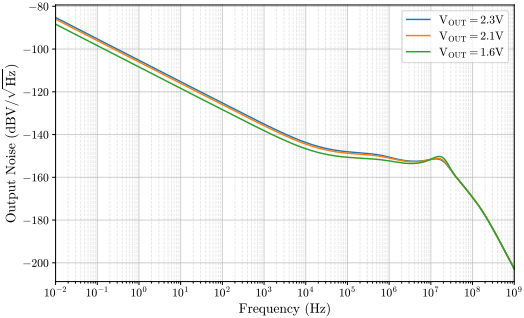


Figure: Output-Referred Noise

Telescopic OTA

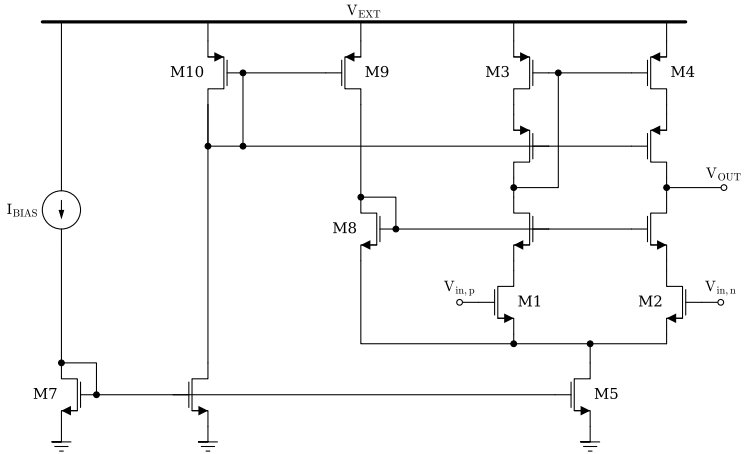


Figure: Telescopic OTA

Telescopic OTA

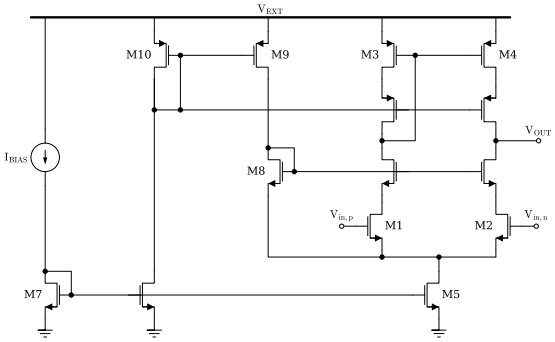


Figure: Telescopic OTA

Leading to the following output swing:

$$V_{REF} - V_T + V_{DSAT} \leq V_{OUT} \leq V_{EXT} - 2V_{DSAT}$$

Both loop gain and output impedance are substantially increased compared to the 5T OTA. More specifically, the loop gain can be given by:

$$A_{TEL} = \frac{g_{m12}}{g_{tel,out}}$$

whereas the output impedance is given by:

$$g_{tel,out} = \frac{1}{g_{ds2}\frac{g_{ds2C}}{g_{m2C}} + g_{ds4}\frac{g_{ds4C}}{g_{m4C}}}$$

Bias transistors should be sized in the following manner in order to ensure maximum output swing:

$$\frac{W_8}{L_8} = \frac{W_{3,4}}{4L_{3,4}}, \quad \frac{W_{10}}{L_{10}} = \frac{W_{1,2}}{4L_{1,2}}$$

Telescopic OTA (Sizing)

- **Input Pair & Active Load:** Large W and L for reducing the Flicker noise and increasing the loop gain.
- **Cascode Transistors:** Small W to ensure saturation, no need for larger transconductance.
- **Tail Current:** Large L for better current mirroring, large W for increasing the bias current and reducing the Thermal noise at higher frequencies.
- **Pass Device:** Large W to improve transconductance and loop gain.
- **Divider:** Small resistors to minimize noise and decrease PD's gate voltage @ No-Load.
- **Compensation:** Nulling resistor and compensation capacitor for improving UGBW

Device	Value / Sizing	Unit
M_1, M_2 (Input Pair)	25.0/5.0	µm
M_{1C}, M_{2C} (NMOS Cascode)	0.65/1.0	µm
M_{3C}, M_{4C} (PMOS Cascode)	2.0/1.0	µm
M_3, M_4 (Active Load)	40.0/5.0	µm
M_8 (NMOS Bias)	0.65/4.0	µm
M_9, M_{10} (PMOS Bias)	2.0/4.0	µm
M_5 (Tail Current)	0.95/1.0	µm
M_6 (Bias Current)	0.28/1.0	µm
M_7 (Input Current)	0.28/1.0	µm
M_{PD} (Pass Device)	40.0/0.13	µm
R_1 (Divider)	22.00	kΩ
R_{21} (Divider)	24.00	kΩ
R_{22} (Divider)	29.33	kΩ
R_{23} (Divider)	66.00	kΩ
R_c (Compensation)	20.00	kΩ
C_c (Compensation)	7.00	pF

Table: Device Sizing for the Telescopic OTA LDO

5T OTA (Simulation Results)

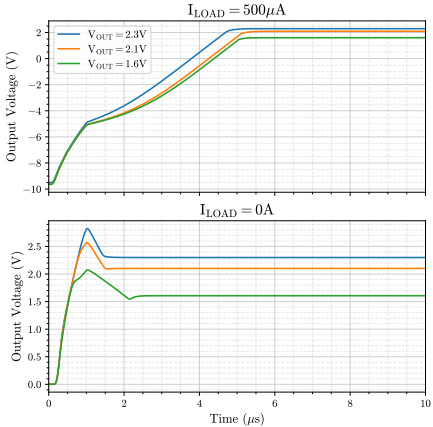


Figure: Ramp Response

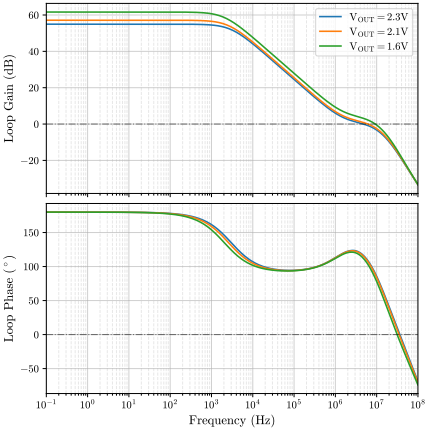


Figure: Bode Plot

5T OTA (Simulation Results)

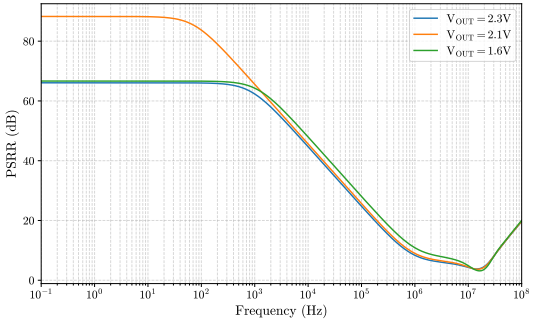


Figure: PSRR

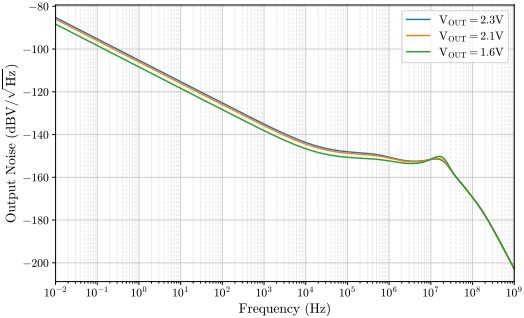


Figure: Output-Referred Noise

Bode Plot

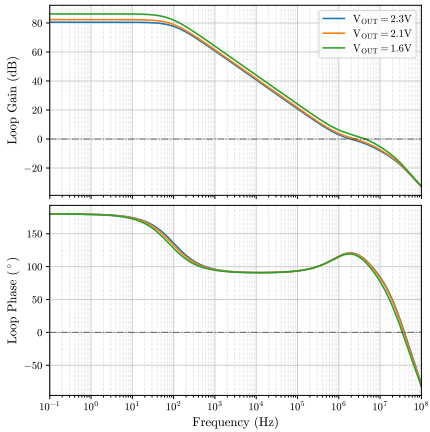


Figure:

Simulation Results

Parameter	Target Specification	5T-OTA (Min – Max)	Telescopic-OTA (Min – Max)
Trim Voltage 1	$= 2.3\,\text{V}$	2.293 – 2.299	2.294
Trim Voltage 2	$= 2.1\,\text{V}$	2.096 – 2.102	2.096
Trim Voltage 3	$= 1.6\,\text{V}$	1.602 – 1.607	1.6
Quiescent Current	$< 100\,\mu\text{A}$	48.16 – 94.49	52.56 – 84.12
Phase Margin ($I_{load} = 0$)	$> 60^\circ$	77.62 – 99.2	74.15 – 99
Phase Margin ($I_{load} = 500\,\mu\text{A}$)	$> 70^\circ$	82.7 – 116.2	107.2 – 120.1
Loop Gain @ 1 Hz	$> 50\,\text{dB}$	54.95 – 61.63	80.53 – 86.23
PSRR @ 1 Hz	$> 65\,\text{dB}$	66.07 – 88.27	70.73 – 75.86
PSRR @ 1 MHz	$> 2\,\text{dB}$	8.43 – 10.95	6.373 – 8.537
PSRR @ 10 MHz	$> 2\,\text{dB}$	4.353 – 4.934	2.107 – 2.118
Output-Referred Noise @ 10 Hz	$< -115\,\text{dBV}/\sqrt{\text{Hz}}$	-118.3 – -115.1	-118.6 – -115.5
Output-Referred Noise @ 1 kHz	$< -125\,\text{dBV}/\sqrt{\text{Hz}}$	-138.1 – -135	-138.1 – -135.1
Output-Referred Noise @ 10 kHz	$< -134\,\text{dBV}/\sqrt{\text{Hz}}$	-146.5 – -143.6	-145.4 – -142.5
Output-Referred Noise @ 100 kHz	$< -143\,\text{dBV}/\sqrt{\text{Hz}}$	-150.6 – -148	-148 – -145.3
Output-Referred Noise @ 1 MHz	$< -155\,\text{dBV}/\sqrt{\text{Hz}}$	-152.3 – -150.5	-149.9 – -148.3

Table: Simulated Performance Summary (Across All Trims)

Conclusion

- The **Telescopic OTA** provided the best overall noise performance.
- The **Current Mirror OTA** offered superior dynamic response at the cost of slight area increase.
- All topologies successfully met the capacitor-less stability criteria across the specified load range.

Thank you!

Questions?